2025
VCE
Specialist Mathematics
Year 12
Trial Examination 2

Detailed Answers

Kilbaha Education

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Section A

ANSWERS

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Section A

Question 1 **Answer C**

If you eat your dinner, then you can have ice-cream or cake.

Consider the three propositions,

Let *d*: 'You eat your dinner', *c*: 'You have cake', *i:* 'You have ice-cream.'

The original statement is $d \to (c \vee i)$

The contrapositive statement is $\neg(c \vee i) \to \neg d \;\Leftrightarrow\; \neg c \wedge \neg i \to \neg d$

If you didn't have ice-cream and didn't have cake then you didn't eat your dinner.

Question 2 **Answer D**

$f(x) = \dfrac{x^2 - a^2}{x - b} = x + b + \dfrac{b^2 - a^2}{x - b}$ has

$y = x + b$ as an oblique asymptote and $x = b$

as vertical asymptote, provided $|b| \neq a$, **A**. is true.

If $b = a,\ f(x) = x + a,\ x \neq a,$

If $b = -a,\ f(x) = x - a,\ x \neq -a$

So no points of infection when $b = \pm a$.

If $|b| = a$ there is a point of discontinuity. **B**. is true.

Solving $f'(x) = \dfrac{x^2 - 2bx + a^2}{(x-b)^2} = 0,$

gives $x = b \pm \sqrt{b^2 - a^2}$.

If $|b| > a,$ there are two solutions, so two turning points. **C**. is true

$f''(x) = \dfrac{-2(a^2 - b^2)}{(x-b)^3} = 0$ has no solution for *x*, so there are no points of inflection, **D**. is false.

Define $f(x) = \dfrac{x^2 - a^2}{x - b}$ *Done*

propFrac$(f(x))$ $\dfrac{-(a^2 - b^2)}{x - b} + x + b$

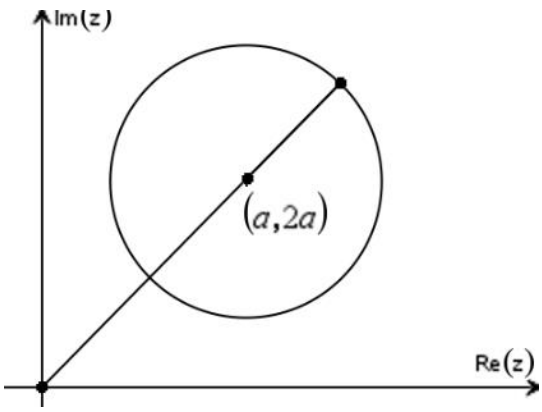
solve$\left(\dfrac{d}{dx}(f(x)) = 0, x\right)$

$x = -\left(\sqrt{b^2 - a^2} - b\right)$ or $x = \sqrt{b^2 - a^2} + b$

$\dfrac{d^2}{dx^2}(f(x))$ $\dfrac{-2 \cdot (a^2 - b^2)}{(x-b)^3}$

Question 3 **Answer A**

The circle $|z - a - 2ai| = r$ has centre at $(a, 2a)$ and radius *r*, where $0 < r < a$, the maximum value of $|z|$ is the furthest point on the circle from the origin, that is the distance of the centre from the origin, which is $\sqrt{(2a)^2 + a^2} = \sqrt{5}a$ plus the radius of the circle, so $|z|_{\max} = \sqrt{5}a + r$



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Question 4 **Answer A**

L_1: $\underset{\sim}{r}(t)=(3+2t)\underset{\sim}{i}+(2-t)\underset{\sim}{j}+(1-3t)\underset{\sim}{k}$ in parametric form is $x=3+2t,\ y=2-t,\ z=1-3t$

Π_1: $x+2y+3z=1,\ \ 3+2t+2(2-t)+3(1-3t)=1,$ solving gives $t=1,\ P_1(5,1,-2)$

Π_2: $x+2y-z=3,\ \ 3+2t+2(2-t)-(1-3t)=3,$ solving gives $t=-1,\ P_2(1,3,4)$

The line intersects each plane at a unique point, respectively.

Question 5 **Answer D**

$\underset{\sim}{a}=4\underset{\sim}{i}+t\underset{\sim}{j}-5\underset{\sim}{k},\qquad \underset{\sim}{b}=-2\underset{\sim}{i}+\underset{\sim}{j}-t\underset{\sim}{k}$

$\dfrac{\underset{\sim}{a}.\underset{\sim}{b}}{|\underset{\sim}{b}|}=2\dfrac{\underset{\sim}{b}.\underset{\sim}{a}}{|\underset{\sim}{a}|}\ \Rightarrow |\underset{\sim}{a}|=2|\underset{\sim}{b}|$

$|\underset{\sim}{a}|=\sqrt{16+t^2+25}=\sqrt{41+t^2},$

$|\underset{\sim}{b}|=\sqrt{4+1+t^2}=\sqrt{5+t^2}$

$|\underset{\sim}{a}|=2|\underset{\sim}{b}|$

$\sqrt{41+t^2}=2\sqrt{5+t^2}$

$41+t^2=4(5+t^2)=20+4t^2$

$3t^2=21,\qquad t^2=7$

$t=\pm\sqrt{7}$

$a:=[4\ \ t\ \ -5]$ $\qquad [4\ \ t\ \ -5]$

$[-2\ \ 1\ \ -t]\rightarrow b$ $\qquad [-2\ \ 1\ \ -t]$

$\text{solve}(\text{norm}(a)=2\cdot\text{norm}(b),t)$

$t=-\sqrt{7}$ or $t=\sqrt{7}$

Question 6 **Answer A**

The equation $y-3=2z,\qquad y-2z=3,\ \ x\in R,$ represents a plane parallel to the x-axis.

Question 7 **Answer B**

$|z-a|=2|z-ai|,\qquad z=x+yi$

$|(x-a)+yi|=2|x+(y-a)i|$

$\sqrt{(x-a)^2+y^2}=2\sqrt{x^2+(y-a)^2}$

$(x-a)^2+y^2=4\left(x^2+(y-a)^2\right)$

$x^2-2xa+a^2+y^2=4x^2+4y^2-8ya+4a^2$

$3x^2+2xa+3y^2-8ya+3a^2=0$

$x^2+\dfrac{2xa}{3}+\dfrac{a^2}{9}+y^2-\dfrac{8ya}{3}+\dfrac{16a^2}{9}=-a^2+\dfrac{a^2}{9}+\dfrac{16a^2}{9}$

$\left(x+\dfrac{a}{3}\right)^2+\left(y-\dfrac{4a}{3}\right)^2=\dfrac{8a^2}{9}$ circle centre $\left(-\dfrac{a}{3},\dfrac{4a}{3}\right)$ radius $\dfrac{2\sqrt{2}a}{3}$

$z:=x+y\cdot i$ $\qquad x+y\cdot i$

$(|z-a|=2\cdot|z-a\cdot i|)^2$

$x^2-2\cdot a\cdot x+y^2+a^2=4\cdot\left(x^2+(y-a)^2\right)$

$\text{completeSquare}\left(x^2-2\cdot a\cdot x+y^2+a^2=4\cdot\left(x^2+(\right.\right.$

$-3\cdot\left(x+\dfrac{a}{3}\right)^2-3\cdot\left(y-\dfrac{4\cdot a}{3}\right)^2=\dfrac{-8\cdot a^2}{3}$

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Question 8 **Answer D**

The plane $\underset{\sim}{r}(s,t)=4\underset{\sim}{i}+\beta\underset{\sim}{j}-\underset{\sim}{k}+t\left(3\underset{\sim}{i}+4\underset{\sim}{j}+2\underset{\sim}{k}\right)+s\left(\underset{\sim}{i}+3\underset{\sim}{j}-\underset{\sim}{k}\right)$

has $\underset{\sim}{v}_1=3\underset{\sim}{i}+4\underset{\sim}{j}+2\underset{\sim}{k}$ and $\underset{\sim}{v}_2=\underset{\sim}{i}+3\underset{\sim}{j}-\underset{\sim}{k}$

as vectors lying in the plane, the vector

$\underset{\sim}{v}_1\times\underset{\sim}{v}_2=-10\underset{\sim}{i}+5\underset{\sim}{j}+5\underset{\sim}{k}=-5\left(2\underset{\sim}{i}-\underset{\sim}{j}-\underset{\sim}{k}\right)$

is parallel to the normal to the plane,

let $\underset{\sim}{n}=2\underset{\sim}{i}-\underset{\sim}{j}-\underset{\sim}{k}$ be the normal to the plane,

$|\underset{\sim}{n}|=\sqrt{4+1+1}=\sqrt{6}$

$v1:=[3\ \ 4\ \ 2]$	$[3\ \ 4\ \ 2]$
$v2:=[1\ \ 3\ \ -1]$	$[1\ \ 3\ \ -1]$
$\text{crossP}(v1,v2)$	$[-10\ \ 5\ \ 5]$
$r:=[x\ \ y\ \ z]$	$[x\ \ y\ \ z]$
$n:=[2\ \ -1\ \ -1]$	$[2\ \ -1\ \ -1]$
$p:=[4\ \ \beta\ \ -1]$	$[4\ \ \beta\ \ -1]$
$\text{dotP}(p,n)=\text{dotP}(r,n)$	$\beta-9=-2\cdot x+y+z$
$\text{solve}\left(\lvert\beta-9\rvert=\sqrt{6}\cdot\text{norm}(n),\beta\right)$	$\beta=3$ or $\beta=15$

A point on the plane is $(4,\beta,-1)$, when $s=0,\ t=0$, the equation of the plane is

$2(x-4)-1(y-\beta)-1(z+1)=0$ or $2x-y-z=9-\beta$

The distance of the plane $ax+by+cz=d$ from the origin is $\dfrac{|d|}{\sqrt{a^2+b^2+c^2}}$

So that $\dfrac{|9-\beta|}{\sqrt{6}}=\sqrt{6},\quad 9-\beta=6,\quad \beta=3,\quad \beta-9=6,\quad \beta=15$

Question 9 **Answer C**

The plane $-2x+y+z=8$

has a normal

$\underset{\sim}{n}=-2\underset{\sim}{i}+\underset{\sim}{j}+\underset{\sim}{k},\quad |\underset{\sim}{n}|=\sqrt{4+1+1}=\sqrt{6}$

The line $\dfrac{x+1}{3}=\dfrac{y-5}{2}=1-z$

or $\dfrac{x+1}{3}=\dfrac{y-5}{2}=\dfrac{z-1}{-1}$

has direction

$\underset{\sim}{v}=3\underset{\sim}{i}+2\underset{\sim}{j}-\underset{\sim}{k},\quad |\underset{\sim}{v}|=\sqrt{9+4+1}=\sqrt{14}$

$\underset{\sim}{v}.\underset{\sim}{n}=-6+2-1=-5$

the angle between the line and the plane is

$$\sin^{-1}\left(\frac{|\underset{\sim}{v}.\underset{\sim}{n}|}{|\underset{\sim}{v}||\underset{\sim}{n}|}\right)=\sin^{-1}\left(\frac{5}{\sqrt{14\times6}}\right)=\sin^{-1}\left(\frac{5\sqrt{21}}{42}\right)$$

$$=\cos^{-1}\left(\frac{\sqrt{1239}}{42}\right)$$

$n:=[-2\ \ 1\ \ 1]$	$[-2\ \ 1\ \ 1]$
$[3\ \ 2\ \ -1]\rightarrow v$	$[3\ \ 2\ \ -1]$
$\lvert\text{dotP}(v,n)\rvert$	5
$\sin^{-1}\left(\dfrac{\lvert\text{dotP}(v,n)\rvert}{\text{norm}(v)\cdot\text{norm}(n)}\right)$	$\sin^{-1}\left(\dfrac{5\cdot\sqrt{21}}{42}\right)$
$\cos^{-1}\left(\dfrac{\sqrt{1239}}{42}\right)$	$\sin^{-1}\left(\dfrac{5\cdot\sqrt{21}}{42}\right)$

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Question 10 **Answer B**

$\frac{dy}{dx} = y\sin(x) = f(x, y)$ using Euler's Method

$y_1 = y_0 + h\, f(x_0, y_0) = y_0 + hy_0 \sin(x_0) = y_0(1 + h\sin(x_0))$

$y_2 = y_1 + h\, f(x_1, y_1) = y_1 + hy_1 \sin(x_1) = y_1(1 + h\sin(x_0 + h))$

$= y_0(1 + h\sin(x_0))(1 + h\sin(x_0 + h))$

Define $f(x,y)=y\cdot\sin(x)$	*Done*
$x1:=x0+h$	$h+x0$
$y1:=y0+h\cdot f(x0,y0)$	$h\cdot\sin(x0)\cdot y0+y0$
$x2:=x1+h$	$2\cdot h+x0$
$y2:=y1+h\cdot f(x1,y1)$	$h\cdot(h\cdot\sin(x0)+1)\cdot\sin(h+x0)\cdot y0+h\cdot\sin(x0)\cdot y0+y0$
factor$(y2)$	$(h\cdot\sin(x0)+1)\cdot(h\cdot\sin(h+x0)+1)\cdot y0$

Question 11 **Answer C**

$V = \frac{\pi}{3} r^2 h, \quad r = r(t), \quad h = h(t)$

Using implicit differentiation and the product rule

$$\frac{dV}{dt} = \frac{\pi}{3}\frac{d}{dt}\left(r^2 h\right) = \frac{\pi}{3}\left(2rh\frac{dr}{dt} + r^2\frac{dh}{dt}\right)$$

Question 12 **Answer C**

When $x = 0,\ y = 0,\ \frac{dy}{dx} = -1$ and when $x = \pm 2\ \ \frac{dy}{dx} = 0$

only satisfied by $\frac{dy}{dx} = \frac{x^2 - 4}{y^2 + 4}$

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Question 13 **Answer A**

$$a = \frac{d\left(\frac{1}{2}v^2\right)}{dx} = \frac{3bx - 2b^2}{x^3}$$

$$\tfrac{1}{2}v^2 = \int \frac{3bx - 2b^2}{x^3}\,dx = \int \left(3bx^{-2} - 2b^2x^{-3}\right)dx$$

$$\tfrac{1}{2}v^2 = -3bx^{-1} + b^2x^{-2} + c$$

Now when $v = 0 \quad x = b$

$$0 = -3 + 1 + c \quad \Rightarrow \quad c = 2$$

$$\tfrac{1}{2}v^2 = \frac{-3b}{x} + \frac{b^2}{x^2} + 2 = \frac{2x^2 - 3bx + b^2}{x^2}$$

$$\tfrac{1}{2}v^2 = \frac{(2x - b)(x - b)}{x^2}$$

So that when $v = 0 \quad x = b$ and $x = \frac{b}{2}$

$$\text{deSolve}\left(v \cdot v' = \frac{3 \cdot b \cdot x - 2 \cdot b^2}{x^3} \text{ and } v(b) = 0, x, v\right)$$

$$v^2 = \frac{2 \cdot \left(2 \cdot x^2 - 3 \cdot b \cdot x + b^2\right)}{x^2}$$

$$\text{solve}\left(\frac{2 \cdot \left(2 \cdot x^2 - 3 \cdot b \cdot x + b^2\right)}{x^2} = 0, x\right)$$

$$x = \frac{b}{2} \text{ or } x = b$$

Question 14 **Answer D**

$$z^4 + a^4 i = 0$$

$$z^4 = -a^4 i = a^4 \operatorname{cis}\left(-\frac{\pi}{2} + 2k\pi\right) \qquad k \in Z$$

$$z = a \operatorname{cis}\left(-\frac{\pi}{8} + \frac{k\pi}{2}\right)$$

so one root when $k = 0$ is $z = a \operatorname{cis}\left(-\frac{\pi}{8}\right)$, which lies on the circle of radius a at an angle of -22.5^0 from the positive real axis, all the other roots must be equally spaced and on the circle of radius a, the roots do not occur in conjugate pairs, so option **D.** is correct.

Question 15 **Answer D**

$V = 20 \text{ m s}^{-1}, \quad \theta = 40^0$

$$H_{\max} = \frac{V^2 \sin^2(\theta)}{2g} = \frac{20^2 \sin^2\left(40^0\right)}{2 \times 9.8} = 8.43 \text{m}$$

$$y = Vt\sin(\theta) - \frac{1}{2}gt^2$$

solving $1.5 = 40t\sin\left(40^0\right) - 4.9t^2$ gives $t = 2.5$ s

$$x = Vt\cos(\theta), \quad x = 40 \times 2.5\cos\left(40^0\right) = 38.32$$

All of Alan, Ben and Colin are all correct.

$\mathbf{v}:=20 \blacktriangleright 20. \quad \boldsymbol{\theta}:=40 \blacktriangleright 40 \quad \mathbf{y}:=1.5 \blacktriangleright 1.5$

$\mathbf{g}:=9.8 \blacktriangleright 9.8$

$$\mathbf{h}:=\frac{\mathbf{v}^2 \cdot (\sin(\boldsymbol{\theta}))^2}{2 \cdot \mathbf{g}} \blacktriangleright 8.4322$$

$$\text{solve}\left(\mathbf{y}=\mathbf{v} \cdot t \cdot \sin(\boldsymbol{\theta}) - \frac{1}{2} \cdot \mathbf{g} \cdot t^2, t\right) | t>0$$

$\blacktriangleright t=0.1224 \text{ or } t=2.5012$

$\mathbf{x}:=\mathbf{v} \cdot t \cdot \cos(\boldsymbol{\theta}) | t=2.5012 \blacktriangleright 38.3206$

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Question 16 **Answer A**

$$\frac{dQ}{dt} = \text{inflow} - \text{outflow} = 2b \times \frac{3}{2} - \frac{bQ}{V}$$

and the volume $V = V(t) = 10 + (2b - b)t$

$$\frac{dQ}{dt} = 3b - \frac{bQ}{10 + bt}, \qquad Q(0) = 2$$

has the solution

$$Q(t) = \frac{3t^2 + 60t + 40}{2(t+10)} \quad \text{when } b = 1.$$

$$\text{deSolve}\left(q' = 3\cdot b - \frac{b\cdot q}{10 + b\cdot t} \text{ and } q(0) = 2, t, q\right)$$

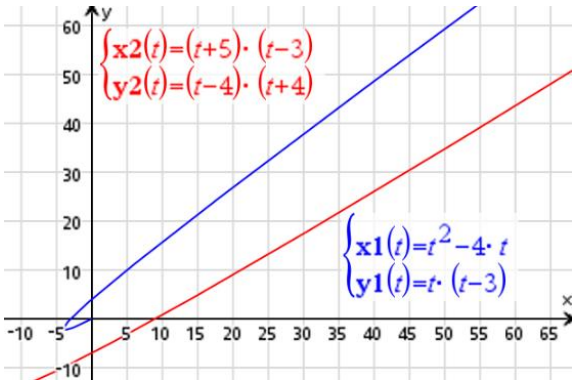
$$q = \frac{3\cdot b^2\cdot t^2 + 60\cdot b\cdot t + 40}{2\cdot(b\cdot t + 10)}$$

$$q = \frac{3\cdot b^2\cdot t^2 + 60\cdot b\cdot t + 40}{2\cdot(b\cdot t + 10)} \,|\, b = 1$$

$$q = \frac{3\cdot t^2 + 60\cdot t + 40}{2\cdot(t + 10)}$$

Question 17 **Answer B**

All of **A. C.** and **D.** are false, **B.** is true.



Define $r1(t) = [x1(t) \;\; y1(t)]$	*Done*
Define $r2(t) = [x2(t) \;\; y2(t)]$	*Done*
Define $s(t) = r1(t) - r2(t)$	*Done*
$\text{norm}(s(t))$	$\sqrt{45\cdot t^2 - 276\cdot t + 481}$
$\text{solve}\left(\frac{d}{dt}(\text{norm}(s(t))) = 0, t\right)$	$t = \frac{46}{15}$
$\text{norm}\left(s\left(\frac{46}{15}\right)\right)$	$\frac{17\cdot\sqrt{5}}{5}$

Question 18 **Answer B**

$$v(t) = \frac{20}{\pi}\arcsin\left(1 - \frac{t}{4}\right)$$

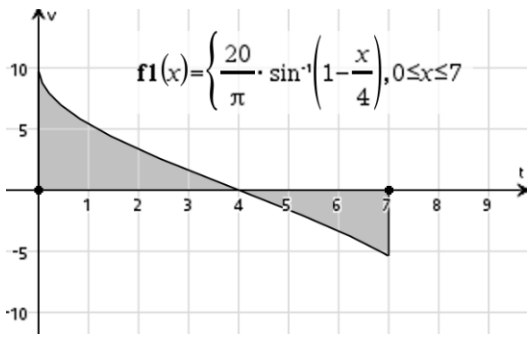
$$s = \int_0^7 |v(t)|\,dt = 22.11$$

$$s = \int_0^4 v(t)\,dt + \left|\int_4^7 v(t)\,dt\right|$$

$$s = 14.535 + 7.575$$

$$s = 22.11$$

$\int_0^4 f1(x)\,dx$	14.5352
$\int_4^7 f1(x)\,dx$	-7.5754
$\int_0^7 \lvert f1(x)\rvert\,dx$	22.1106



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Question 19 **Answer C**

$\mu = 162, \quad n = 36, \quad \bar{x} = 160, \quad s = 8.5, \qquad CI: \ \bar{x} \pm z \times \dfrac{s}{\sqrt{n}}$

80% $z = 1.2816$, $(158.18, 161.82)$ does not contain μ

84% $z = 1.4051$ $(158.01, 161.99)$ does not contain μ

85% $z = 1.4395$ $(157.96, 162.04)$ does contain μ

90% $z = 1.6449$ $(157.67, 162.33)$

zInterval 8.5,160,36,0.84: *stat.results*

"Title"	"z Interval"
"CLower"	158.01
"CUpper"	161.99
"x̄"	160.00
"ME"	1.99
"n"	36.00
"σ"	8.50

zInterval 8.5,160,36,0.85: *stat.results*

"Title"	"z Interval"
"CLower"	157.96
"CUpper"	162.04
"x̄"	160.00
"ME"	2.04
"n"	36.00
"σ"	8.50

Question 20 **Answer B**

$X \overset{d}{=} N\left(\mu, \sigma^2\right), \qquad \bar{X} \overset{d}{=} N\left(\mu, \dfrac{\sigma^2}{n}\right)$

$\left(\dfrac{\text{invNorm}(0.002)}{\text{invNorm}(0.02)}\right)^2$ 1.9640

When $n = n_1$, $\Pr\left(\bar{X} < T\right) = 0.02$ $\qquad \dfrac{T - \mu}{\dfrac{\sigma}{\sqrt{n_1}}} = -2.0537$

When $n = n_2$, $\Pr\left(\bar{X} > T\right) = 0.998$, $\Pr\left(\bar{X} < T\right) = 0.002$ $\qquad \dfrac{T - \mu}{\dfrac{\sigma}{\sqrt{n_2}}} = -2.8782$

$\sqrt{\dfrac{n_2}{n_1}} = \dfrac{2.8782}{2.0537}, \qquad \dfrac{n_2}{n_1} = (1.4041)^2 = 1.96 \approx 2$

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Section B

Question 1

a.i.	$f(x)=\dfrac{2x^3-6x^2-20x+48}{x^3+x^2-10x+8}$ $f(x)=\dfrac{2(x-4)(x-2)(x+3)}{(x-2)(x-1)(x+4)}$ $f(x)=\dfrac{2(x-4)(x+3)}{(x-1)(x+4)} \quad x\neq 2$ $f(x)=2-\dfrac{8(x+2)}{(x-1)(x+4)} \quad x\neq 2$ The straight line asymptotes are $y=2,\quad x=1,\quad x=-4$	Define $p(x)=2\cdot x^3-6\cdot x^2-20\cdot x+48$ *Done* factor$(p(x))$ $2\cdot(x-4)\cdot(x-2)\cdot(x+3)$ Define $q(x)=x^3+x^2-10\cdot x+8$ *Done* factor$(q(x))$ $(x-2)\cdot(x-1)\cdot(x+4)$ Define $f1(x)=\dfrac{p(x)}{q(x)}$ *Done* propFrac$(f1(x))$ $2-\dfrac{8\cdot(x+2)}{x^2+3\cdot x-4}$	A1
ii.	The graph crosses the y-axis when $x=0,\ y=6,\ (0,6)$ The graph crosses the x-axis when $2(x-4)(x+3)=0,\ x=4,-3,$ $(4,0),(-3,0)$ Note the point at $x=2,\ f(2)=-\dfrac{10}{3}$ is a point of discontinuity. The graph crosses the horizontal asymptote $y=2$ at $x=-2,\ (-2,2)$	$f1(x)$ $\dfrac{2\cdot(x^2-x-12)}{x^2+3\cdot x-4}$ factor$(f1(x))$ $\dfrac{2\cdot(x-4)\cdot(x+3)}{(x-1)\cdot(x+4)}$ $f1(0)$ 6 $\lim\limits_{x\to 2}(f1(x))$ $\dfrac{-10}{3}$	A1
iii.	$f'(x)=\dfrac{8(x^2+4x+10)}{(x-1)^2(x+4)^2}$ for turning points $f'(x)=0$ but solving $x^2+4x+10=(x+2)^2+6=0$ has no real solutions, so that $f'(x)>0$, the gradient is always positive, so there are no turning points.	$\dfrac{d}{dx}(f1(x))$ $\dfrac{8\cdot(x^2+4\cdot x+10)}{(x^2+3\cdot x-4)^2}$ solve$\left(\dfrac{d}{dx}(f1(x))=0,x\right)$ false	A1

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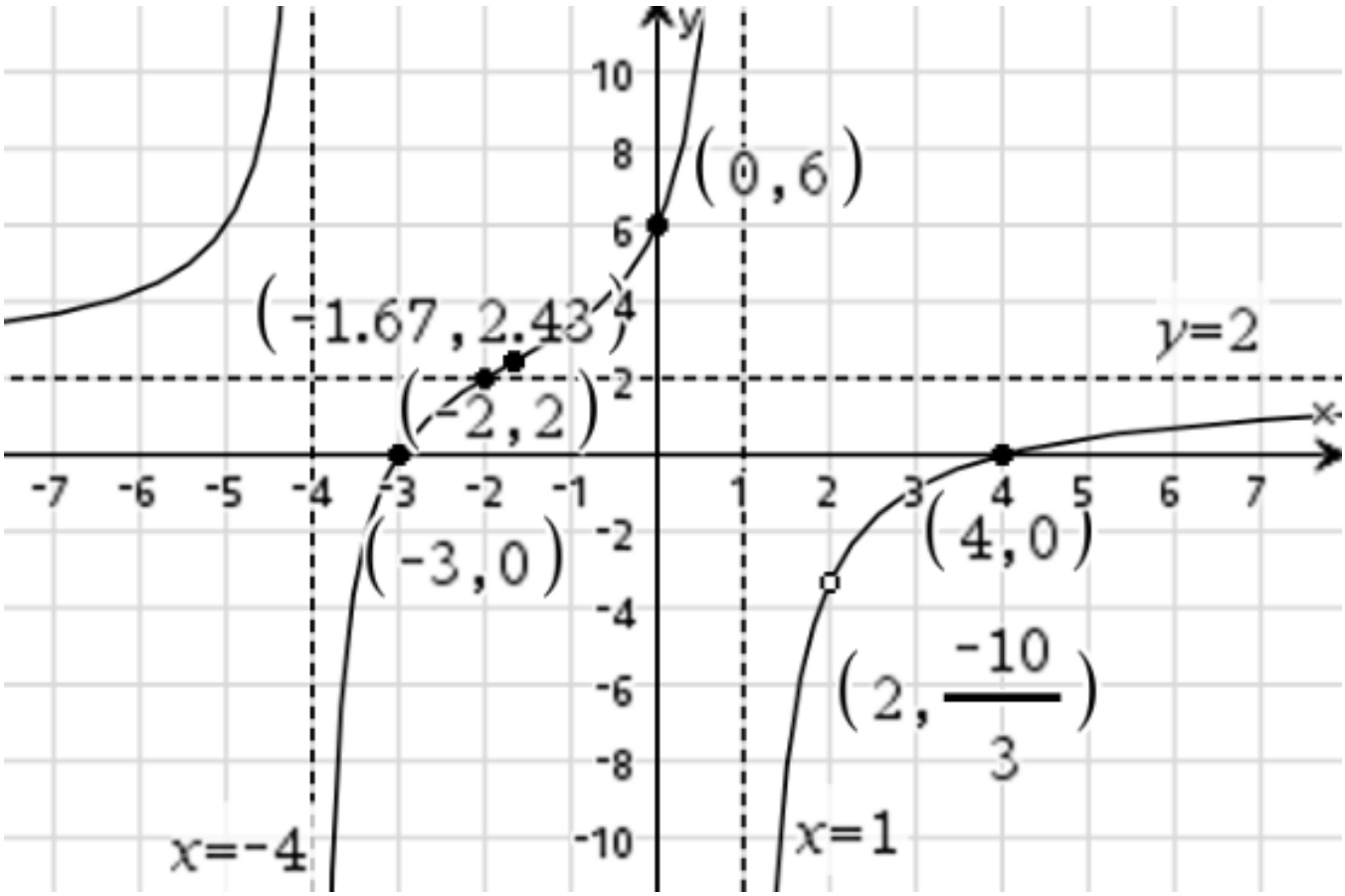
iv.	The point of inflexion occurs when $f''(x)=0$ at $(-1.67, 2.43)$	solve$\left(\frac{d^2}{dx^2}(f1(x))=0,x\right)$ $x=-1.6687$ $f1(-1.6687)$ 2.4260	A1
	Must have open circle at $\left(2,-\frac{10}{3}\right)$ as the point of discontinuity and label the point of inflexion at $(-1.67, 2.43)$ and crossing the horizontal asymptote at $(-2,2)$, all axial intercepts $(4,0)$, $(-3,0)$, $(0,6)$ and all asymptotes clearly labelled.		A1

G1



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b.i	$g(x)=\dfrac{2x^3-6x^2-20x+48}{x^3-(3+a)x^2+(3a+2)x-2a}$ $g(x)=\dfrac{2(x-4)(x-2)(x+3)}{(x-2)(x-1)(x-a)}$ $g(x)=\dfrac{2(x-4)(x+3)}{(x-1)(x-a)} \quad x\neq 2$ When $a=1$ there are only two straight line asymptotes, that is $y=2$ and $x=1$. Note when $a=-3$ or $a=4$ then we get two points of discontinuity and one vertical asymptote and one horizontal asymptote.	propFrac$(g(x))$ $\dfrac{2\cdot(a\cdot x-a-12)}{(x-1)\cdot(x-a)}+2$ factor$(x^3-(3+a)\cdot x^2+(3\cdot a+2)\cdot x-2\cdot a)$ $(x-2)\cdot(x-1)\cdot(x-a)$ Define $g(x)=\dfrac{p(x)}{(x-2)\cdot(x-1)\cdot(x-a)}$ *Done* factor$(g(x))$ $\dfrac{2\cdot(x-4)\cdot(x+3)}{(x-1)\cdot(x-a)}$	A1
ii.	$y=2$ and $x=1$ are always straight line asymptotes. If $a\in R\setminus\{1,-3,4\}$ there will be another vertical asymptote.	$(x-1)(x-a)=x^2-(1+a)x+a$ must have two solutions, $\Delta=(1+a)^2-4a=a^2-2a+1=(a-1)^2>0$ $a\neq 1$	A1
iii.	The graph will cross the horizontal asymptote $y=2$ at $x=\dfrac{a+12}{a}$ provided that $a\neq 0$, so $a\in R\setminus\{-3,0,4\}$	propFrac$(g(x))$ $\dfrac{2\cdot(a\cdot x-a-12)}{(x-1)\cdot(x-a)}+2$	A1
iv.	$g'(x)=\dfrac{-2(ax^2-2(a+12)x+13a+12)}{(x-1)^2(x-a)^2}$ for two turning points $g'(x)=0$ solving $ax^2-2(a+12)x+13a+12=0 \quad a\neq 0$ $\Delta=(-2(a+12))^2-4a(13a+12)$ $\Delta=-48a^2+48a+576$ $\Delta>0,\ -3<a<4$ but $a\neq 0,1$ so for two turning points $a\in(-3,0)\cup(0,1)\cup(1,4)$ or $a\in(-3,4)\setminus\{0,1\}$	$\dfrac{d}{dx}(g(x))$ $\dfrac{-2\cdot(a\cdot x^2-2\cdot(a+12)\cdot x+13\cdot a+12)}{(x-1)^2\cdot(x-a)^2}$ $delta:=(-2\cdot(a+12))^2-4\cdot a\cdot(13\cdot a+12)$ $-48\cdot a^2+48\cdot a+576$ solve$(delta>0,a)$ $-3<a<4$	M1 A1

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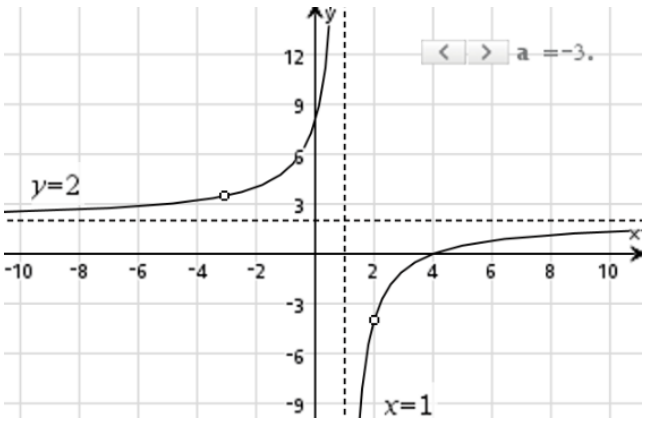

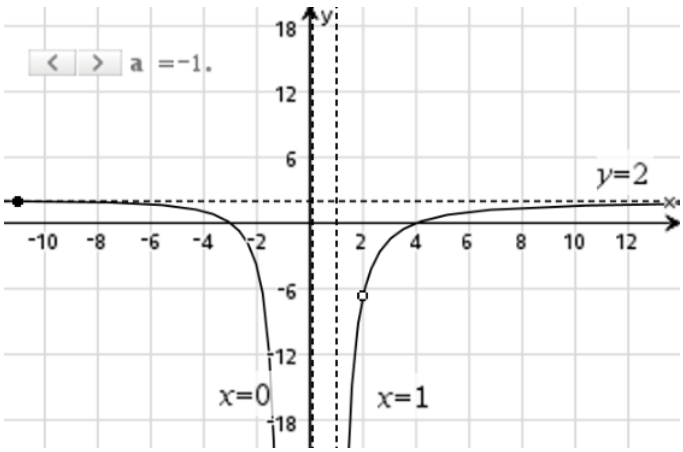

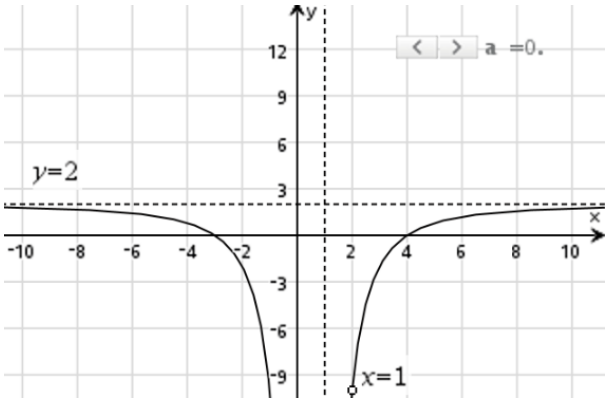

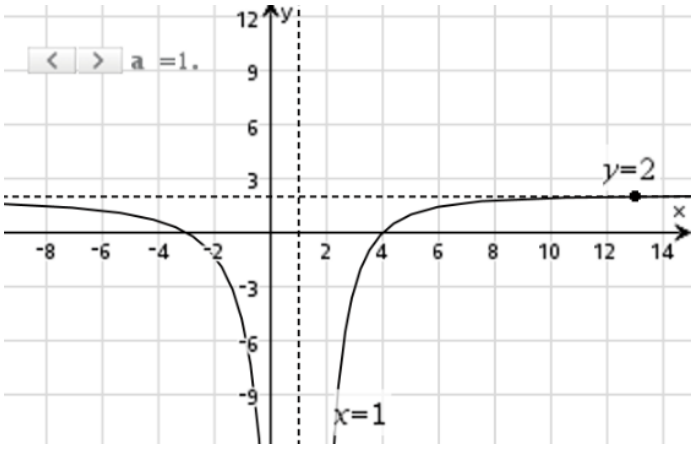

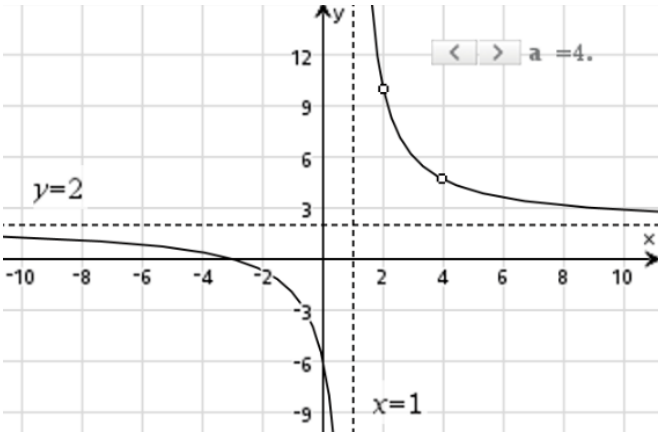

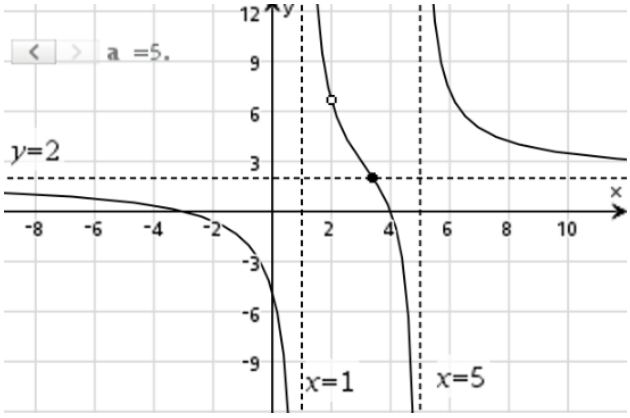

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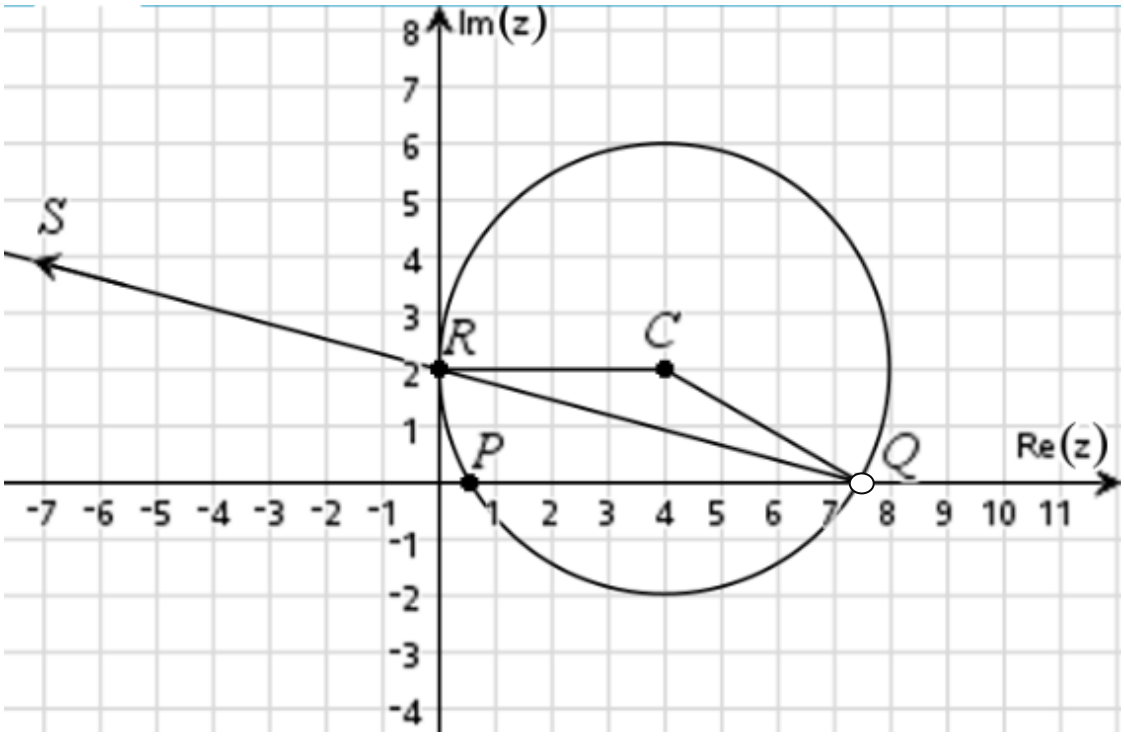
Question 2

a.i.	$u=4+2i,\quad z=x+yi$ $\lvert z-u\rvert=4$ $\lvert(x-4)+(y-2)i\rvert=4$ $\sqrt{(x-4)^2+(y-2)^2}=4$ $(x-4)^2+(y-2)^2=16$ A circle centre at $C(4,2)$ radius 4.	$u:=4+2\cdot i$ $\quad 4+2\cdot i$ $z:=x+y\cdot i$ $\quad x+y\cdot i$ $\lvert z-u\rvert=4$ $\quad \sqrt{x^2-8\cdot x+y^2-4\cdot y+20}=4$ $(\lvert z-u\rvert=4)^2$ $\quad x^2-8\cdot x+y^2-4\cdot y+20=16$ completeSquare$(x^2-8\cdot x+y^2-4\cdot y+20=16,\{x$ $(x-4)^2+(y-2)^2=16$	A1
ii.	crosses the x-axis, $y=0$ $(x-4)^2+4=16$ $x=4\pm\sqrt{12}$ $x_P=4-2\sqrt{3},\ x_Q=4+2\sqrt{3}$ $P(4-2\sqrt{3},0)\quad Q(4+2\sqrt{3},0)$ crosses the y-axis, $x=0$ $4^2+(y-2)^2=16$ $y=2$ $R(0,2)$	solve$((x-4)^2+(y-2)^2=16,x)\lvert y=0$ $x=4-2\cdot\sqrt{3}$ or $x=2\cdot\sqrt{3}+4$ solve$((x-4)^2+(y-2)^2=16,y)\lvert x=0$ $\quad y=2$	A1
iii.			G1

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b.i.	gradient $m(QR)=\dfrac{-2}{4+2\sqrt{3}}=-2+\sqrt{3}$ $\tan^{-1}\left(-2+\sqrt{3}\right)=-\dfrac{\pi}{12},\quad z_Q=4+2\sqrt{3}+0i$ but the angle the ray makes with the positive real axis is $\pi-\dfrac{\pi}{12}=\dfrac{11\pi}{12}$ S: $\text{Arg}\left(z-\left(4+2\sqrt{3}\right)\right)=\dfrac{11\pi}{12},\quad k=\dfrac{11}{12}$ $y=\left(\sqrt{3}-2\right)x+2,\quad x<4+2\sqrt{3}$	$zq:=4-2\cdot\sqrt{3}$ $\quad 4-2\cdot\sqrt{3}$ $\text{angle}(z-zq)=\dfrac{11\cdot\pi}{12}$ $\dfrac{\pi\cdot\text{sign}(y)}{2}-\tan^{-1}\left(\dfrac{x+2\cdot\left(\sqrt{3}-2\right)}{y}\right)=\dfrac{11\cdot\pi}{12}$ $\text{expand}\left(\text{solve}\left(\dfrac{\pi\cdot\text{sign}(y)}{2}-\tan^{-1}\left(\dfrac{x+2\cdot\left(\sqrt{3}-2\right)}{y}\right)\right.\right.$ $x+2\cdot\left(\sqrt{3}-2\right)<0$ and $y=\left(\sqrt{3}-2\right)\cdot\left(x+2\cdot\left(\sqrt{3}-2\right)\right.$	A1 A1
ii.	$C(4,2)\quad d(QC)=d(RC)=4$ $d(RQ)=\sqrt{\left(2\sqrt{3}+4\right)^2+2^2}=\sqrt{32+16\sqrt{3}}$ Using the cosine rule to find angle RCQ $\angle RCQ=\theta=\cos^{-1}\left(\dfrac{4^2+4^2-\left(32+16\sqrt{3}\right)}{2\times4\times4}\right)=\cos^{-1}\left(-\dfrac{\sqrt{3}}{2}\right)=\dfrac{5\pi}{6}$ $A=\dfrac{1}{2}r^2\left(\theta-\sin(\theta)\right)=\dfrac{1}{2}\times4^2\left(\dfrac{5\pi}{6}-\sin\left(\dfrac{5\pi}{6}\right)\right)=8\left(\dfrac{5\pi}{6}-\dfrac{1}{2}\right)$ $A=\dfrac{20\pi}{3}-4=\dfrac{4(5\pi-3)}{3}\ \text{unit}^2$		M1 A1
iii.	Consider a line PT perpendicular to the ray QR, it has a gradient of $m(PT)=\dfrac{4+2\sqrt{3}}{2}=2+\sqrt{3}$, the equation of the line PT is $y2=\left(2+\sqrt{3}\right)\left(x-4+2\sqrt{3}\right)$ this line intersects the ray $y1=\left(\sqrt{3}-2\right)x+2,\quad x<4+2\sqrt{3}$ at M which is the midpoint of P and T, solving gives $x_M=1,\quad y_M=\sqrt{3}$ $\dfrac{4-2\sqrt{3}+x_T}{2}=1,\quad \dfrac{y_T+0}{2}=\sqrt{3}$ $x_T=-2+2\sqrt{3},\quad y_T=2\sqrt{3}$ $z_T=-2+2\sqrt{3}+2\sqrt{3}i$	Define $y1=\sqrt{3}\cdot x-2\cdot x+2$ $\quad$ *Done* Define $y2=\left(2+\sqrt{3}\right)\cdot\left(x-4+2\cdot\sqrt{3}\right)$ $\quad$ *Done* $\text{solve}(y1=y2,x)$ $\quad x=1$ $y1\|x=1$ $\quad \sqrt{3}$ $y2\|x=1$ $\quad \sqrt{3}$	M1 A1

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c.	$p(z)=z^5+\alpha z^4+\beta z^3+\chi z^2+\delta z-200,$ by the conjugate root theorem, $\bar{u}=4-2i$ and $\frac{\bar{u}}{2}=2-i$ are also roots, $u+\bar{u}=8,\ u.\bar{u}=16-4i^2=20$ so that $z^2-8z+20$ is one quadratic factor, $\frac{1}{2}(u+\bar{u})=4,\quad \frac{1}{4}u.\bar{u}=4-i^2=5$ so that z^2-4z+5 is another quadratic factor, expanding $(z^2-8z+20)(z^2-4z+5)$ $=z^4-12z^3+57z^2-120z+100$ so $(z-2)$ is a linear factor, expanding $p(z)=(z-2)(z^2-8z+20)(z^2-4z+5)$ $=z^5-14z^4+81z^3-234z^2+340z-200$ $\alpha=-14,\ \beta=81,\ \chi=-234,\quad \delta=340$	$(z-(4-2\cdot i))\cdot(z-(4+2\cdot i))\cdot(z-(2-i))\cdot(z-(2+i))$ $(z^2-8\cdot z+20)\cdot(z^2-4\cdot z+5)$ expand$((z^2-8\cdot z+20)\cdot(z^2-4\cdot z+5))$ $z^4-12\cdot z^3+57\cdot z^2-120\cdot z+100$ expand$((z-2)\cdot(z^4-12\cdot z^3+57\cdot z^2-120\cdot z+10$ $z^5-14\cdot z^4+81\cdot z^3-234\cdot z^2+340\cdot z-200$ cSolve$(z^5-14\cdot z^4+81\cdot z^3-234\cdot z^2+340\cdot z-2$ $z=4+2\cdot i$ or $z=4-2\cdot i$ or $z=2+i$ or $z=2-i$ or $z=$	A1

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Question 3

a.	$\cos(3A)=\cos(2A+A)$ $=\cos(2A)\cos(A)-\sin(2A)\sin(A)$ $=(2\cos^2(A)-1)\cos(A)-2\sin^2(A)\cos(A)$ $=\cos(A)(2\cos^2(A)-1)-2\cos(A)(1-\cos^2(A))$ $=2\cos^3(A)-\cos(A)-2\cos(A)+2\cos^3(A)$ $=4\cos^3(A)-3\cos(A)$ $\underset{\sim}{r}(t)=\cos(t)\underset{\sim}{i}+\cos(3t)\underset{\sim}{j}$ $x=\cos(t),\quad y=\cos(3t)$ $y=4x^3-3x,\quad x\in[0,1]$	M1 A1
b.i.	$\sin(3A)=\sin(2A+A)$ $=\sin(2A)\cos(A)+\cos(2A)\sin(A)$ $=2\cos^2(A)\sin(A)+(1-2\sin^2(A))\sin(A)$ $=2\sin(A)(1-\sin^2(A))+\sin(A)(1-2\sin^2(A))$ $=2\sin(A)-2\sin^3(A)+\sin(A)-2\sin^3(A)$ $=3\sin(A)-4\sin^3(A)$ $x=\cos(t),\qquad y=\cos(3t)$ $\frac{dx}{dt}=-\sin(t),\quad \frac{dy}{dt}=-3\sin(3t)$ $\frac{dy}{dx}=\frac{dy}{dt}.\frac{dx}{dt}=\frac{-3\sin(3t)}{-\sin(t)}=\frac{3(3\sin(t)-4\sin^3(t))}{\sin(t)}=\frac{3\sin(t)(3-4\sin^2(t))}{\sin(t)}$ $\frac{dy}{dx}=9-12\sin^2(t)$	M1 A1
ii.	for turning points, $\frac{dy}{dx}=0 \Rightarrow 9-12\sin^2(t)=0,\quad \sin^2(t)=\frac{3}{4}$ $\sin(t)=\pm\frac{\sqrt{3}}{2}$ but $0\le t\le\frac{\pi}{2},\ 0\le\sin(t)\le 1,\quad t=\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{3}$ $x\left(\frac{\pi}{3}\right)=\cos\left(\frac{\pi}{3}\right)=\frac{1}{2},\quad y\left(\frac{\pi}{3}\right)=\cos(\pi)=-1$ the turning point is at $\left(\frac{1}{2},-1\right)$	A1

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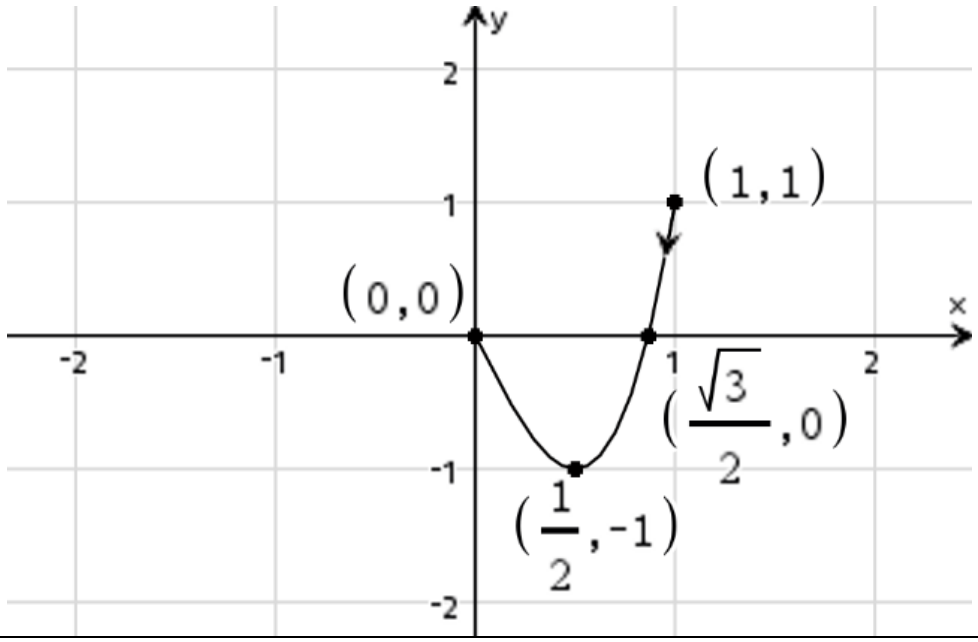
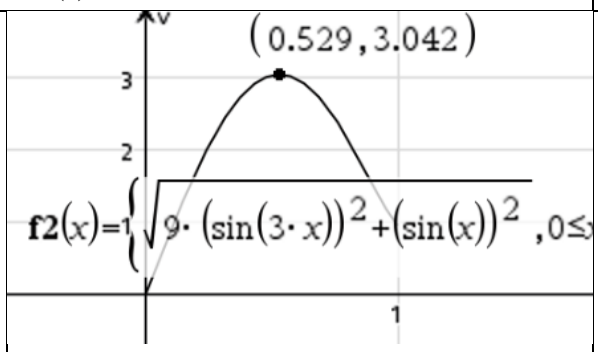
iii.	$t=0,\ (1,1),\ \ t=\frac{\pi}{6},\left(\frac{\sqrt{3}}{2},0\right),\ \ t=\frac{\pi}{3},\left(\frac{1}{2},-1\right),\ \ t=\frac{\pi}{2},(0,0)$ shape and endpoints 	G2
d.i.	$\dot{\underset{\sim}{r}}(t)=\underset{\sim}{v}(t)=-\sin(t)\underset{\sim}{i}-3\sin(3t)\underset{\sim}{j}$ $\lvert\underset{\sim}{v}(t)\rvert^2=(-\sin(t))^2+(-3\sin(3t))^2=\sin^2(t)+9\sin^2(3t)=\sin^2(t)+9(3\sin(t)-4\sin^3(t))^2$ $\lvert\underset{\sim}{v}(t)\rvert^2=\sin^2(t)+9(9\sin^2(t)-24\sin^4(t)+16\sin^6(t))$ $\lvert\underset{\sim}{v}(t)\rvert=\sqrt{82\sin^2(t)-216\sin^4(t)+144\sin^6(t)}$ $\lvert\underset{\sim}{v}(t)\rvert=\sqrt{\sin^2(t)(82-216\sin^2(t)+144\sin^4(t))}=\lvert\sin(t)\rvert\sqrt{82-216\sin^2(t)+144\sin^4(t)}$ $0\le t\le\frac{\pi}{2}$ so the modulus is not needed, speed $\lvert\underset{\sim}{v}(t)\rvert=\sin(t)\sqrt{82-216\sin^2(t)+144\sin^4(t)}$	M1 A1
ii.	Solving for maximum speed $\frac{d}{dt}(\lvert\underset{\sim}{v}(t)\rvert)=0$ Since $0\le t\le\frac{\pi}{2}$ gives $t=0.529$ $v_{\max}=\lvert\underset{\sim}{v}(0.529)\rvert=3.042$ or graphically. 	A1
e.	$L=\int_0^{\frac{\pi}{2}}\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt=\int_0^{\frac{\pi}{2}}\lvert\underset{\sim}{v}(t)\rvert\,dt=\int_0^{\frac{\pi}{2}}\sin(t)\sqrt{a+b\sin^2(t)+c\sin^4(t)}\,dt$ $a=82,\ \ b=-216,\ \ c=144$	A1

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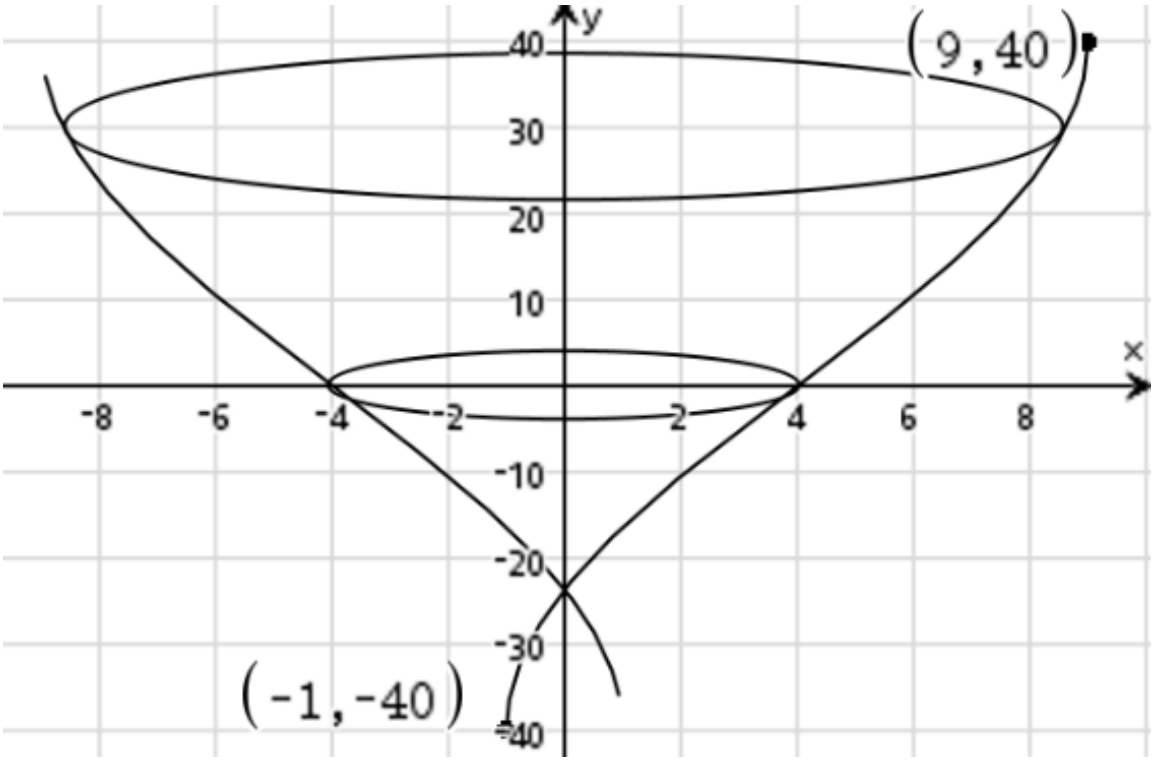
Question 4

a.	$f(x)=\frac{80}{\pi}\sin^{-1}\left(\frac{x-4}{5}\right)$ domain is $[-1,9]$ and the range is $[-40,40]$	Define $f1(x)=\frac{80}{\pi}\cdot\sin^{-1}\left(\frac{x-4}{5}\right)$ *Done* domain$(f1(x),x)$ $-1\le x\le 9$ $f1(-1)$ -40 $f1(9)$ 40	A1
			A1
b.	The diameter of the bowl at the top is 17.239 cm.	solve$(f1(x)=30,x)$ $x=8.619$ $2\cdot 8.619397$ 17.239	A1

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c.	$y=\frac{80}{\pi}\sin^{-1}\left(\frac{x-4}{5}\right)$, $\frac{\pi y}{80}=\sin^{-1}\left(\frac{x-4}{5}\right)$, $\sin\left(\frac{\pi y}{80}\right)=\frac{x-4}{5}$, $x=4+5\sin\left(\frac{\pi y}{80}\right)$ $V_x=\pi\int_a^b x^2\,dy$ $V=\pi\int_0^H\left(4+5\sin\left(\frac{\pi y}{80}\right)\right)^2 dy$	A1
d.	$S_y=2\pi\int_{y_1}^{y_2} x(y)\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$ $x(y)=4+5\sin\left(\frac{\pi y}{80}\right)$, $\frac{dx}{dy}=\frac{\pi}{16}\cos\left(\frac{\pi y}{80}\right)$ $S_y=2\pi\int_0^{30}\left(4+5\sin\left(\frac{\pi y}{80}\right)\right)\sqrt{1+\frac{\pi^2}{256}\cos^2\left(\frac{\pi y}{80}\right)}\,dy$	A1
e.	$\frac{dV}{dt}=-k\sqrt{h}$, $k>0$ $\frac{dV}{dh}=\pi\left(4+5\sin\left(\frac{\pi h}{80}\right)\right)^2$ $\frac{dt}{dh}=\frac{dt}{dV}\frac{dV}{dh}$ $\frac{dt}{dh}=\frac{\pi\left(4+5\sin\left(\frac{\pi h}{80}\right)\right)^2}{-k\sqrt{h}}$ $T=\int_{25}^{0}\frac{\pi\left(4+5\sin\left(\frac{\pi h}{80}\right)\right)^2}{-k\sqrt{h}}\,dh=60\times 10$ Solving gives $k=1.688$ solve$\left(\int_{25}^{0}\frac{\pi\cdot\left(4+5\cdot\sin\left(\frac{\pi\cdot h}{80}\right)\right)^2}{-k\cdot\sqrt{h}}\,dh=10\cdot 6\right.$ $k=1.688$	A1 M1 A1

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Question 5 $O(0,0,0)$ $P(3,-2,1)$ and $Q(q,-1,3)$

a. $\overrightarrow{OP} = 3\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}, \quad \overrightarrow{OQ} = q\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k}$

$\overrightarrow{OP} \times \overrightarrow{OQ} = -5\underset{\sim}{i} + (q-9)\underset{\sim}{j} + (2q-3)\underset{\sim}{k}$

$A_{OPQ} = \frac{1}{2}\left|\overrightarrow{OP} \times \overrightarrow{OQ}\right| = \frac{1}{2}\sqrt{5\left(q^2 - 6q + 23\right)} = \frac{5\sqrt{6}}{2}$

$q = -1,\ 7$ A1

$vop := [3 \quad -2 \quad 1]$	$[3 \quad -2 \quad 1]$
$[q \quad -1 \quad 3] \rightarrow voq$	$[q \quad -1 \quad 3]$
$\text{crossP}(vop, voq)$	$[-5 \quad q-9 \quad 2\cdot q-3]$
$\text{norm}(\text{crossP}(vop, voq))$	$\sqrt{5\cdot\left(q^2-6\cdot q+23\right)}$
$\text{solve}(\text{norm}([-5 \quad q-9 \quad 2\cdot q-3])=5\cdot\sqrt{6}, q)$	$q=-1 \text{ or } q=7$

b. Let $R(x_0, y_0, z_0)$ be the point on the plane $5x + 7y - z = 8$, which is closest to the point Q.

Now $\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ} = (x_0 - q)\underset{\sim}{i} + (y_0 + 1)\underset{\sim}{j} + (z_0 - 3)\underset{\sim}{k}$ is parallel to the normal to the plane

$\underset{\sim}{n} = 5\underset{\sim}{i} + 7\underset{\sim}{j} - \underset{\sim}{k}$, so that $\overrightarrow{QR} = \lambda \underset{\sim}{n}$ and M1

$\left|\overrightarrow{QR}\right| = \frac{\sqrt{3}}{5} = \left|\lambda \underset{\sim}{n}\right| = \left|\lambda\sqrt{25+49+1}\right| = \sqrt{75}\left|\lambda\right| = 5\sqrt{3}\left|\lambda\right| \ \Rightarrow\ \lambda = \pm\frac{1}{25}$

Now, $x_0 - q = 5\lambda, \quad y_0 + 1 = 7\lambda, \quad z_0 - 3 = -\lambda,$ but $R(x_0, y_0, z_0)$

lies on the plane, so that $5x_0 + 7y_0 - z_0 = 5(q + 5\lambda) + 7(7\lambda - 1) - (3 - \lambda) = 8$

$\lambda = \frac{1}{25}, \quad q = 3, \quad \lambda = -\frac{1}{25}, \quad q = \frac{21}{5}$ A1

$[5 \quad 7 \quad -1] \rightarrow von$	$[5 \quad 7 \quad -1]$
$\text{norm}(von)$	$5\cdot\sqrt{3}$
$5\cdot(q+5\cdot\lambda)+7\cdot(7\cdot\lambda-1)-(3-\lambda)=8$	$5\cdot q+75\cdot\lambda-10=8$
$\text{solve}(5\cdot q+75\cdot\lambda-10=8, q) \mid \lambda=\frac{1}{25}$	$q=3$
$\text{solve}(5\cdot q+75\cdot\lambda-10=8, q) \mid \lambda=\frac{-1}{25}$	$q=\frac{21}{5}$

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Alternatively

$5x+7y-z=8=d$, $\underset{\sim}{n}=5\underset{\sim}{i}+7\underset{\sim}{j}-\underset{\sim}{k}$, $\overrightarrow{OQ}=q\underset{\sim}{i}-\underset{\sim}{j}+3\underset{\sim}{k}$

$$\frac{\left|\underset{\sim}{n}.\overrightarrow{OQ}-d\right|}{\left|\underset{\sim}{n}\right|}=\left|\frac{5q-7-3-8}{\sqrt{25+49+1}}\right|=\frac{\sqrt{3}}{5}$$ M1

$|5q-18|=3$

$5q-18=3, \qquad 18-5q=3$

$q=\frac{21}{5},\ 3$ A1

Another Method

Let R be the point on the plane $5x+7y-z=8$, which is closest to the point Q.

Take a convenient point S on the plane, say $S(0,0,-8)$, now

$\overrightarrow{QS}=\overrightarrow{OS}-\overrightarrow{OQ}=-q\underset{\sim}{i}+\underset{\sim}{j}-11\underset{\sim}{k}$ and the normal to the plane is $\underset{\sim}{n}=5\underset{\sim}{i}+7\underset{\sim}{j}-\underset{\sim}{k}$ and

$|\underset{\sim}{n}|=\sqrt{25+49+1}=\sqrt{75}=5\sqrt{3}$, $\hat{\underset{\sim}{n}}=\frac{1}{5\sqrt{3}}\left(5\underset{\sim}{i}+7\underset{\sim}{j}-\underset{\sim}{k}\right)$, the distance is M1

$\left|\overrightarrow{QS}.\hat{\underset{\sim}{n}}\right|=\frac{|-5q+7+11|}{5\sqrt{3}}=\frac{|18-5q|}{5\sqrt{3}}=\frac{\sqrt{3}}{5}$ so that $|18-5q|=3$

$18-5q=3, \qquad 18-5q=-3$

$q=3,\ \frac{21}{5}$ A1

c. The line has a direction $\underset{\sim}{v}=\overrightarrow{PQ}=\overrightarrow{OQ}-\overrightarrow{OP}=(q-3)\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}$ the line through P is

$\underset{\sim}{r}(t)=\overrightarrow{OP}+t\overrightarrow{PQ}=3\underset{\sim}{i}-2\underset{\sim}{j}+\underset{\sim}{k}+t\left((q-3)\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}\right)$ A1

$\underset{\sim}{r}(t)=\overrightarrow{OP}+t\overrightarrow{PQ}=\left(3+t(q-3)\right)\underset{\sim}{i}+(-2+t)\underset{\sim}{j}+(1+2t)\underset{\sim}{k}$

Note that many alternative answers are possible.

d. The parametric equation of the line is $x=3+t(q-3)$, $y=-2+t$, $z=1+2t$ this intersects the plane $5x+7y-z=10$, so that $5\left(3+t(q-3)\right)+7(-2+t)-(1+2t)=10$ or $(5q-10)t=10$ (1) but also $x=3+t(q-3)=4$ (2), solving (1) and (2) gives

$q=4,\ t=1,\ by=-1,\ bz=3,\ B(4,-1,3)$ A1

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e. The two lines, $L_1:\ \underset{\sim}{r}_1(t)=(3+t(q-3))\underset{\sim}{i}+(-2+t)\underset{\sim}{j}+(1+2t)\underset{\sim}{k}$

$L_2:\underset{\sim}{r}_2(s)=(2+2s)\underset{\sim}{i}+(-1+s)\underset{\sim}{j}+(3+2s)\underset{\sim}{k}$ are parallel, line (1)

has direction $\underset{\sim}{v}_1=(q-3)\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}$ and line (2) has direction

$\underset{\sim}{v}_2=2\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k},\quad q-3=2,\quad q=5$

Let the point $S(x_1,y_1,z_1)$ be on line (2) and is the closest to the point

$P(3,-2,1)$ on line (1). $\overrightarrow{PS}=\overrightarrow{OS}-\overrightarrow{OP}=(x_1-3)\underset{\sim}{i}+(y_1+2)\underset{\sim}{j}+(z_1-1)\underset{\sim}{k}$ M1

is perpendicular to the direction of the lines, $\underset{\sim}{v}=2\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}$ so that

$\overrightarrow{PS}.\underset{\sim}{v}=2(x_1-3)+(y_1+2)+2(z_1-1)=0\ (3)$ but $S(x_1,y_1,z_1)$ lies on

line (2) so that $x_1=2+2s,\quad y_1=-1+s,\quad z_1=3+2s\ (4)$ solving (3) and (4) gives $s=-\frac{1}{3}$

and $\overrightarrow{PS}=\frac{1}{3}(-5\underset{\sim}{i}+2\underset{\sim}{j}+4\underset{\sim}{k})$, the closest distance between the two lines is

$\left|\overrightarrow{PS}\right|=\frac{1}{3}\sqrt{25+4+16}=\frac{\sqrt{45}}{3}=\sqrt{5}$ A1

$x1:=2+2\cdot s$	$2\cdot s+2$
$y1:=-1+s$	$s-1$
$z1:=3+2\cdot s$	$2\cdot s+3$
$\text{solve}(2\cdot(x1-3)+y1+2+2\cdot(z1-1)=0,s)$	$s=\frac{-1}{3}$

$vor:=[x1\ \ y1\ \ z1]$	$[2\cdot s+2\ \ s-1\ \ 2\cdot s+3]$
$\text{norm}(vor-vop)\|s=\frac{-1}{3}$	$\sqrt{5}$

Alternatively

Take a point $P_1(3,-2,1)$ on line L_1 and a point $P_2(2,-1,3)$ on line L_2, both lines have

direction $\underset{\sim}{v}=2\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}$, and $|\underset{\sim}{v}|=\sqrt{4+1+4}=3,\quad \hat{\underset{\sim}{v}}=\frac{1}{3}(2\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k})$ M1

$q-3=2,\quad q=5$

$\overrightarrow{P_1P_2}=\overrightarrow{OP_2}-\overrightarrow{OP_1}=-\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}$, the required distance is

$\left|\overrightarrow{P_1P_2}\times\hat{\underset{\sim}{v}}\right|=\frac{1}{3}(6\underset{\sim}{j}-3\underset{\sim}{k})=\left|2\underset{\sim}{j}-\underset{\sim}{k}\right|=\sqrt{4+1}=\sqrt{5}$ A1

$[3\ \ -2\ \ 1]\rightarrow p1$	$[3\ \ -2\ \ 1]$
$[2\ \ -1\ \ 3]\rightarrow p2$	$[2\ \ -1\ \ 3]$
$[2\ \ 1\ \ 2]\rightarrow v$	$[2\ \ 1\ \ 2]$
$\text{crossP}(p2-p1,v)$	$[0\ \ 6\ \ -3]$
$\text{norm}(\text{crossP}(p2-p1,\text{unitV}(v)))$	$\sqrt{5}$

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f. Take a point $P_1(3,-2,1)$ on line L_1 the direction of line L_1 is $\underset{\sim}{v}_1=(q-3)\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}$ and let W be the point on the line L_1 which is closest to the point $A(-1,2,-2)$.

$\overrightarrow{AP_1}=\overrightarrow{OP_1}-\overrightarrow{OA}=4\underset{\sim}{i}-4\underset{\sim}{j}+3\underset{\sim}{k}$ M1

Now the required distance is $\left|\overrightarrow{AW}\right|=\left|\overrightarrow{AP_1}\times\hat{\underset{\sim}{v}}_1\right|=\sqrt{35}$

solving gives $q=\dfrac{2}{5}$, 4 A1

$[-1\ \ 2\ \ -2]\rightarrow pa$ $\qquad [-1\ \ 2\ \ -2]$

$[q-3\ \ 1\ \ 2]\rightarrow v1$ $\qquad [q-3\ \ 1\ \ 2]$

$pap1:=p1-pa$ $\qquad [4\ \ -4\ \ 3]$

$\text{crossP}(pap1,v1)$ $\qquad [-11\ \ 3\cdot q-17\ \ 4\cdot q-8]$

$\text{solve}(\text{norm}(\text{crossP}(pap1,\text{unitV}(v1)))=\sqrt{35},q)$

$q=\dfrac{2}{5}$ or $q=4$

Alternatively

Let $W(x_2,y_2,z_2)$ be the point on the line L_1: $\underset{\sim}{r}_1(t)=(3+t(q-3))\underset{\sim}{i}+(-2+t)\underset{\sim}{j}+(1+2t)\underset{\sim}{k}$, which is closest to the point $A(-1,2,-2)$, Now

$\overrightarrow{AW}=\overrightarrow{OW}-\overrightarrow{OA}=(x_2+1)\underset{\sim}{i}+(y_2-2)\underset{\sim}{j}+(z_2+2)\underset{\sim}{k}$ is perpendicular to

$\underset{\sim}{v}_1=(q-3)\underset{\sim}{i}+\underset{\sim}{j}+2\underset{\sim}{k}$ the direction of the line, so that

$\underset{\sim}{v}_1=(q-3)(x_q+1)+(y_2-2)+2(z_2+2)=0 \quad (5)$ M1

but since the closest distance from W to the point A is

$\left|\overrightarrow{AW}\right|=\sqrt{(x_2+1)^2+(y_2-2)^2+(z_2+2)^2}=\sqrt{35}$

or $(x_2+1)^2+(y_2-2)^2+(z_2+2)^2=35 \quad (6)$ and the point W lies on the line

$L_1,\quad x_2=(3+t(q-3)),\quad y_2=-2+t,\quad z_2=1+2t \quad (7)$, solving (5) (6) and (7)

gives $q=\dfrac{2}{5}$, 4 A1

$x2:=3+t\cdot(q-3)$ $\qquad (q-3)\cdot t+3$

$y2:=-2+t$ $\qquad t-2$

$z2:=1+2\cdot t$ $\qquad 2\cdot t+1$

$(q-3)\cdot(x2+1)+y2-2+2\cdot(z2+2)=0$

$(q^2-6\cdot q+14)\cdot t+4\cdot q-10=0$

$(x2+1)^2+(y2-2)^2+(z2+2)^2=35$

$(q^2-6\cdot q+14)\cdot t^2+(8\cdot q-20)\cdot t+41=35$

$\text{solve}((q^2-6\cdot q+14)\cdot t+4\cdot q-10=0 \text{ and } (q^2-6\cdot$

$q=\dfrac{2}{5}$ and $t=\dfrac{5}{7}$ or $q=4$ and $t=-1$

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Question 6

a.i	$B(7.58, 5.75)$	Define $f1(x)=\sec\left(\frac{3\cdot x}{16}\right)-1$ *Done* Define $f2(x)=\frac{x^2}{10}$ *Done* solve$(f1(x)=f2(x),x)\|5<x<10$ $x=7.5849$ $f2(7.584)$ 5.7517	A1
ii.	$V=\pi\int_0^{7.58}\left((f(x))^2-(g(x))^2\right)dx$ $V=\pi\int_0^{7.58}\left(\frac{x^4}{100}-\left(\sec\left(\frac{3x}{16}\right)-1\right)^2\right)dx$ $V=109.05\ \text{cm}^3$	$xb:=7.5849013047902$ 7.5849 $\pi\cdot\int_0^{xb}\left((f2(x))^2-(f1(x))^2\right)dx$ 109.0525	A1
b.	bananas $B\overset{d}{=}N(118, 6^2)$ total of n independent bananas $T=B_1+B_2+.....+B_n$ $\text{E}(T)=n\text{E}(B)=118n$ $\text{Var}(T)=n\text{Var}(B)=36n,$ $T\overset{d}{=}N(118n, 36n)$ $\Pr(T<1500)=0.058$ $n=13$	solve$\left(\frac{118\cdot n-1500}{6\cdot\sqrt{n}}=\text{invNorm}(0.058,0,1),n\right)$ $n=12.4301$ normCdf$(-\infty,1500,118\cdot n,6\cdot\sqrt{n})\|n=13$ 0.0580	A1
c.i.	$H_0:\ \mu=118$ $H_1:\ \mu\neq118$ two-sided test		A1

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ii.	$n=30,\quad \bar{B}\overset{d}{=}N\left(118,\frac{6^2}{30}\right),$ $p\text{ value}=2\Pr\left(\bar{B}<116\mid\mu=118\right)$ $=0.0679$ Since the p value $0.0679>0.05$ there is no evidence to support the alternative hypothesis, yes accept the null hypothesis.	$2\cdot\text{normCdf}\left(-\infty,116,118,\frac{6}{\sqrt{30}}\right)$ 0.0679 zTest 118,6,116,30,0: *stat.results* "Title" "z Test" "Alternate Hyp" "μ ≠ μ0" "z" -1.8257 "PVal" 0.0679 "x̄" 116.0000 "n" 30.0000 "σ" 6.0000	A1 A1
d.i.	$n=30,\quad \bar{B}\overset{d}{=}N\left(118,\frac{6^2}{30}\right)$ $\Pr\left(\bar{B}<c1\right)=0.025$ $c1=115.853$ $\Pr\left(\bar{B}<c2\right)=0.975$ $c2=120.147$ $\bar{B}_T\overset{d}{=}N\left(116.5,\frac{6^2}{30}\right)$ $\beta=\Pr\left(c1<\bar{B}_T<c2\right)=0.722$	$c1:=\text{invNorm}\left(0.025,118,\frac{6}{\sqrt{30}}\right)$ 115.8530 $c2:=\text{invNorm}\left(0.975,118,\frac{6}{\sqrt{30}}\right)$ 120.1470 $\text{normCdf}\left(c1,c2,116.5,\frac{6}{\sqrt{30}}\right)$ 0.7222	A1 A1
ii.	Using a slider for n, the minimum value of n for which the Type II error $\beta<0.05$ is when $n=208$	$\alpha:=0.05$ $s:=6$ n =208. $\mu0:=118$ ▸ 118 $\mu1:=116.5$ ▸ 116.5000 $c1:=\text{invNorm}\left(\frac{\alpha}{2},\mu0,\frac{s}{\sqrt{n}}\right)$ ▸ 117.1846 $c2:=\text{invNorm}\left(1-\frac{\alpha}{2},\mu0,\frac{s}{\sqrt{n}}\right)$ ▸ 118.8154 $\beta:=\text{normCdf}\left(c1,c2,\mu1,\frac{s}{\sqrt{n}}\right)$ ▸ 0.0499	M1 A1

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e.	bananas $B \overset{d}{=} N\left(118, 6^2\right)$ skin (peel) $S \overset{d}{=} N\left(15, 0.75^2\right)$ editable $T = B - S$ $\mathrm{E}(T) = \mathrm{E}(B) - \mathrm{E}(S) = 118 - 15 = 103$ $\mathrm{Var}(T) = \mathrm{Var}(B) + \mathrm{Var}(S)$ $= 6^2 + 0.75^2$ $= 36.5625$ $T \overset{d}{=} N(103, 36.5625)$ Jack eats $J = T_1 + T_2$ $\mathrm{E}(J) = 2\mathrm{E}(T) = 206$ $\mathrm{Var}(J) = 2 \times 36.5625 = 73.125$ $J \overset{d}{=} N(206, 73.125)$ $\Pr(J > 200) = 0.7586$	$\mathrm{normCdf}\left(200, \infty, 206, \sqrt{73.125}\right)$ 0.7586	A1

End of detailed answers for the
2025 Kilbaha VCE Specialist Mathematics Trial Examination 2

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